Jean-Charles **Quillet**

Senior Software Engineer · Functional Programming
13, rue du clos romain, 35120 Dol-de-Bretagne, FRANCE
(+33) 6 65 13 50 75 • jeancharles.quillet@gmail.com • jeancharles.quillet.org • jecaro • jeancharlesquillet

Summary

Senior software engineer with twenty years of experience across backend systems, network servers, and simulation software. Strong expertise in C++ (10+ years) and functional programming with Haskell (5 years). Proficient in systems design, performance optimization, and developing correct, safe, and robust software. Actively involved in open-source software, both as a contributor and as a maintainer.

Skills

Systems	Distributed systems, backend services, simulation software
Methods	Test-driven development, code review, CI/CD pipelines
Domains	Networking, Computer Graphics, Geographic Information Systems
Programming Languages	Haskell, Rust, C/C++, Zig, Python, Elm, TypeScript, Bash, Lua, Matlab,
Tools	Git, Nix, PostgreSQL, MySQL, Redis, Kafka, Docker, Kubernetes, OpenTelemetry,
Languages	French: native, English: fluent, Spanish: beginner

Experience

CircuitHub — London, UK / Remote

Software engineer — 08/2026 - Present

- Working on a software platform for PCB manufacturing and assembly
- Technical stack: Linux, Haskell, Nix, Docker, PostgresSQL, Nomad

Obsidian Systems — New York, USA / Remote

Software engineer — 01/2026 - 06/2026

- Worked on an automated market maker (AMM) on the Canton Network
- Driving code quality initiatives: set up CI pipelines, automated tests, and a local testing environment for development and end-to-end validation
- Technical stack: Linux, Haskell, Nix, Docker, Canton Network, Daml

Tiko — Zürich, Switzerland / Remote

Backend developer — 01/2024 - 12/2025

- Worked on the interface between IoT devices (over 300k deployed) and backend services
- Implemented observability features for an in-house database, improving monitoring, logging, and performance insights
- Participated in on-call rotation for production monitoring and incident handling
- Technical stack: Linux, Haskell, C++, Java, Nix, Kubernetes, PostgreSQL, MySQL, Redis, Kafka, MQTT, OpenTelemetry

Secucloud — Hamburg, Germany / Remote

Cloud software developer — 11/2020 - 11/2023

- Developed cloud-based network security software as part of a SASE (Secure Access Service Edge) system
- Implemented an Active Directory connector for seamless synchronization of users and groups
- Optimized network proxy performance, increasing throughput by 50%
- Technical stack: Linux, Haskell, TypeScript, Nix, MySQL, Ansible, Redis, Clickhouse

Epitech — Rennes, France

Functional programming teacher — 09/2021 - 06/2023 (part-time)

- Supported students with their Haskell projects
- Technical stack: Linux, Haskell, nix

Scalian — Rennes, France

Software developer — 04/2007 - 11/2020

- Developed simulation software within the Simulation, HPC and Virtual Reality department
- Delivered multiple fixed-price projects on time and within budget for clients including the French MoD, ONERA, CEA, CNES, IRSN
- Architecture and technical lead
- Collaborated with researchers and physicists to gather requirements and deliver tailored software solutions
- Technical stack: Windows/Linux, C++, Python, Matlab, Qt, openscenegraph, gdal

LaBRI: Analysis and indexing video team (University lab) — Bordeaux, France

Research engineer — 07/2006 – 08/2006

- Prepared the video indexing contests TRECVID and ARGOS
- System administration
- Technical stack: Linux, C/C++, XML, Perl, Bash

INRIA: IPARLA team (French IT lab) — Bordeaux, France

Research engineer — 09/2004 – 04/2006

- Visualization of city models on mobile devices through non-photorealistic rendering
- Wrote and presented a scientific paper to the Web3D 2006 conference
- Technical stack: Linux, Windows, Windows Mobile, C/C++, Bash, Perl, VRML97, X3D

Keyghost — Christchurch, New Zealand

Software developer — 04/2003 - 12/2003

- Developed a desktop application to interact with a hardware keylogger
- Technical stack: Windows, Delphi, C/C++, Octave

Visualpix — Bordeaux, France

Internship followed by a software engineer contract — 04/2002 - 12/2002

- Improved a video compression codec by implementing a spatial index
- Technical stack: Linux, C

Glisshouse — Marseille, France

Backend developer, system administrator — 07/2000 - 03/2001

- Web: Maintained an online shop, developed a CMS
- System administration: Installed, set up and maintained the LAN
- Technical stack: Linux, Php, MySQL, Apache, Samba, cron

Education

2001 - 2002	**DESS Multimedia and Information Technology** image synthesis and analysis, 3D geometry and rendering, sound, video	**Bordeaux university**
1999 - 2000	**Master in computer science**, image synthesis and analysis	**Poitiers university**
1998 - 1999	**B.S. in computer science**	**Limoges university**
1996 - 1997	**DEUG computer science and mathematics**	**Limoges university**
1995 - 1996	**Baccalauréat Science**	**Lycée Jean Giraudoux, Châteauroux**

Extracurricular Activity

Sports	Rock climbing, bike commuting, surf, trekking
Culture	Literature (classical, travel, adventure), cinema, attending rock concerts, guitar playing
Community commitments	Member of the steering committee of the rock climbing club of Rennes (2012 to 2016) Volunteered at concert venues of Rennes: Antipode (2011 to 2015), Le Jardin Moderne (2009 to 2010) Volunteered to the music festivals Epipapu (2009, 2010) and la Route du Rock (2008 - 2018)
Travel	New-Zealand, Australia, USA, Canada, Malaysia, Bali, Bolivia, Iceland, Norway, Scotland, Faroe Islands